

Full Stack Web Development Roadmap (2026)

1. Computer Basics

Before coding, learn:

- How internet works
- Browser, server, database
- HTTP & HTTPS
- DNS
- Hosting
- Command Line (Terminal)
- Git & GitHub

Learn:

- HTML basics
 - CSS basics
 - JavaScript basics
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2. Frontend Development

Frontend = What users see in browser.

HTML

Learn:

- Tags
- Forms
- Tables
- Semantic HTML
- Accessibility

CSS

Learn:

- Flexbox
- Grid
- Responsive Design
- Animations
- Media Queries

JavaScript

Learn:

- Variables
 - Functions
 - Arrays
 - Objects
 - DOM
 - Events
 - ES6
 - Async/Await
 - Fetch API
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3. Frontend Framework

Learn React.js

Most popular frontend framework.

Topics:

- Components
- Props
- State
- Hooks
- Routing
- API Calls
- Forms

- Context API

Then Learn:

- Next.js
 - Tailwind CSS
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4. Backend Development

Backend = Server side logic.

Learn Node.js

Topics:

- Express.js
 - REST API
 - Authentication
 - JWT
 - Middleware
 - File Upload
 - Error Handling
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5. Database

SQL Database

Learn:

- MySQL or PostgreSQL

Topics:

- CRUD
- Joins
- Relations

NoSQL Database

Learn:

- MongoDB

Topics:

- Collections
 - Documents
 - Aggregation
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6. Authentication & Security

Learn:

- Login/Register
 - JWT
 - OAuth
 - Password Hashing
 - HTTPS
 - CORS
 - Rate Limiting
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7. APIs

Learn:

- REST API
- GraphQL basics
- Third-party APIs

Projects:

- Weather App
- Payment Gateway
- Maps API

8. Version Control

Git & GitHub

Commands:

```
git init  
git add .  
git commit  
git push
```

Learn:

- Branching
 - Pull Requests
 - Merge
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9. Deployment

Learn Hosting:

Frontend

- [Vercel](#)
- [Netlify](#)

Backend

- [Render](#)
- [Railway](#)

Database Hosting

- [MongoDB Atlas](#)

- [Supabase](#)
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10. Advanced Full Stack

Learn:

- Docker
 - CI/CD
 - Redis
 - WebSockets
 - Microservices
 - Serverless
 - TypeScript
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11. AI + Full Stack (Very Important)

Since you are interested in AI/ML:

Learn:

- Python
- FastAPI
- AI APIs
- LangChain
- Vector Database
- RAG
- OpenAI APIs

Useful:

- [FastAPI](#)
 - [OpenAI Platform](#)
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12. Full Stack Projects

Beginner

- Portfolio Website
- Calculator
- Todo App
- Weather App

Intermediate

- Blog Website
- Chat Application
- E-commerce Website
- Video Streaming Site

Advanced

- AI Chatbot
 - Veterinary Emergency Platform
 - IoT Dashboard
 - Thermal Camera Monitoring System
 - LoRa Monitoring Network
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13. Best Tech Stack (2026)

MERN Stack

- MongoDB
- Express.js
- React.js
- Node.js

AI Stack

- React + FastAPI + PostgreSQL + Docker
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14. Daily Learning Plan

First 3 Months

- HTML
- CSS
- JavaScript
- Git

Next 3 Months

- React
- Node.js
- MongoDB

Next 3 Months

- Full Projects
- Deployment
- APIs

Next 3 Months

- AI Integration
 - Docker
 - Advanced Projects
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15. Best Learning Websites

- [freeCodeCamp](https://www.freecodecamp.org/)
 - [MDN Web Docs](https://developer.mozilla.org/en-US/docs/Web)
 - [W3Schools](https://www.w3schools.com/)
 - [Frontend Mentor](https://www.frontendmentor.net/)
 - [The Odin Project](https://theodinproject.com/)
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16. Tools You Must Install

Code Editor

- [Visual Studio Code](#)

Runtime

- [Node.js](#)

Version Control

- [Git](#)
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17. Final Goal

After completing roadmap you can become:

- Full Stack Developer
- AI Web Developer
- Backend Developer
- Frontend Developer
- DevOps Engineer
- AI Engineer
- Computer Vision Engineer

Your best path based on your interests:

AI + IoT + Full Stack

Build projects using:

- Raspberry Pi
- ESP32
- LoRa
- Thermal Cameras
- AI Dashboard
- Real-time Monitoring

- Emergency Alert Systems

This combination has very high demand in 2026.